

PrimeFocus

July 2026



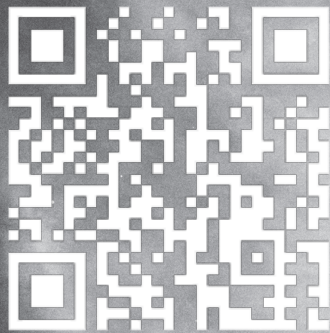
WHEN

July 17, 2026
Doors open at 7:00pm
Meeting starts at 7:30pm
Lecture starts at 8:00pm

WHERE

Unitarian Church
1893 North Vasco Road
Livermore, CA 945571
and via Zoom

TVS QR CODE



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THIS MONTH'S SPEAKER

The Vera C. Rubin Observatory: A New Eye Opens on the Universe **Prof Christopher Stubbs**

The Rubin Observatory is making the transition from construction into full operation. This facility will provide a decade-long time-lapse movie of the Southern sky in 6 passbands. The unprecedented combination of aperture (8.4 m primary, f/1.2 beam) and field of view (~ 10 square degrees) makes Rubin a powerful discovery machine. Professor Stubbs will briefly review the project's history and describe how co-added and difference images can be used for projects ranging from searches for potentially hazardous asteroids to observational cosmology.



Rubin Observatory/NOIRLab/NSF/AURA/Y. AISayyad

Christopher Stubbs joined Harvard in 2003 as a Professor of Physics and of Astronomy. His research interests lie at the intersection of cosmology, particle physics, and gravitation. His research career started with experimental tests of gravitation, performing precision measurements to explore possible modifications to gravity. He was on the faculty at UCSB and was then a member of the University of Washington faculty for a decade before moving to Harvard in 2003. He served as chair of Harvard's Physics Department from 2007 to 2010 and as Dean of Science from 2018 to 2024.

Stubbs is a Fellow of the American Physical Society, a recipient of the National Academy of Sciences Award for Initiative in Research, the NASA Achievement Medal, and a co-recipient (with other members of the High- z Supernova Team who discovered the accelerating expansion of the Universe) of the Gruber Foundation Cosmology Prize and the Breakthrough Prize in Fundamental Physics. Stubbs was awarded a McDonnell Centennial Fellowship and a Packard Fellowship and currently serves on the advisory panel for the Packard Foundation. He is a member of the JASON advisory group, in connection with his interest in arms control and technical aspects of international security issues.

Stubbs was the inaugural Project Scientist for the LSST (now Rubin) sky survey telescope project and remains deeply involved in that effort.

Link to presentation: <https://www.youtube.com/watch?v=LPpymOMSxV0>

NEWS AND NOTES

Meeting Dates

Club Meeting	Board Meeting	PrimeFocus Deadline
July 17	July 20	July 5
August 21	August 24	August 8
September 18	September 21	September 5

Money Matters

As of the last Treasurer’s Report on 6/22/26, our club’s account balance is \$52,242.60, this includes \$14,205.46 in the H2O Rebuild fund.

TVS Welcomes New Members

TVS welcomes new members Dave Bullock, Varghese George, Kalyan Uppalapati, Fatemeh Miri and Wenbo Yuani. Please say hello and chat with them during our meetings.

2026 TVS Club Star Party Schedule

Save the dates for the 2025 Club Star Parties. Del Valle star parties are also public outreach events. They are jointly hosted with the EBRPD and held at the Arroyo Staging Area (Coords: 37.6196638, -121.7528899). The public is invited for the first 1.5-2 hours, while club members can stay the remainder of the night.

July 25: Club/public star party at Del Valle Arroyo Road Staging Area. Set-up 8:00pm, Observing from 8:30-10:00pm.

August 15: Club/public star party at Del Valle Arroyo Road Staging Area. Set-up 8:00pm, Observing from 8:30-10:00pm.

August 22: Club/public star party at Del Valle Arroyo Road Staging Area. Set-up 8:00pm, Observing from 8:30-10:00pm.

Tesla Vintners star parties are open to only club members and their guests. These star parties end at midnight, but participants can leave earlier, should they wish.

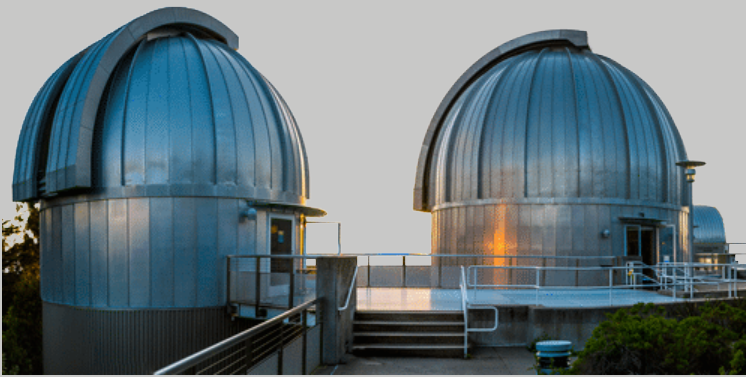
August 8: Tesla Vintner’s Star Party, 5143 Tesla Rd., Livermore. Set-up at 7:30pm, Observing 8:00-Midnight.

October 10: Tesla Vintner’s Star Party, 5143 Tesla Rd., Livermore. Set-up at 6:30pm, Observing 7:00pm-11-30pm.

H2O Open House star parties are open to only club members and their guests. The open house ends at midnight, and all participants are encouraged to stay the duration. The drive to H2O takes about 1 hour, and the caravan leaves promptly from the corner of Mines and Tesla Rds. No gas stations are available on the route, so be prepared. Admission is \$3/car-bring exact change. H2O is a primitive site with two porta-potties. Bring water, food, and warm clothing, as needed. Red flashlights are to be used so observers can preserve their night vision.

September 5: H2O Open House, 5:00pm caravan to H2O PROMPTLY leaves the corner of Mines and Tesla Rds., Livermore. Observing until 11:30pm. Meeting times are tentative.

CALENDAR OF EVENTS



July 17, 18, 24,25, 31 August 1, 7, 8,14, 15, 7:30-10:30pm	
What	Free Telescope Viewing
Who	Chabot Staff
Where	Chabot Space and Science Center, 10000 Skyline Blvd. Oakland, CA 94619
Cost	Free

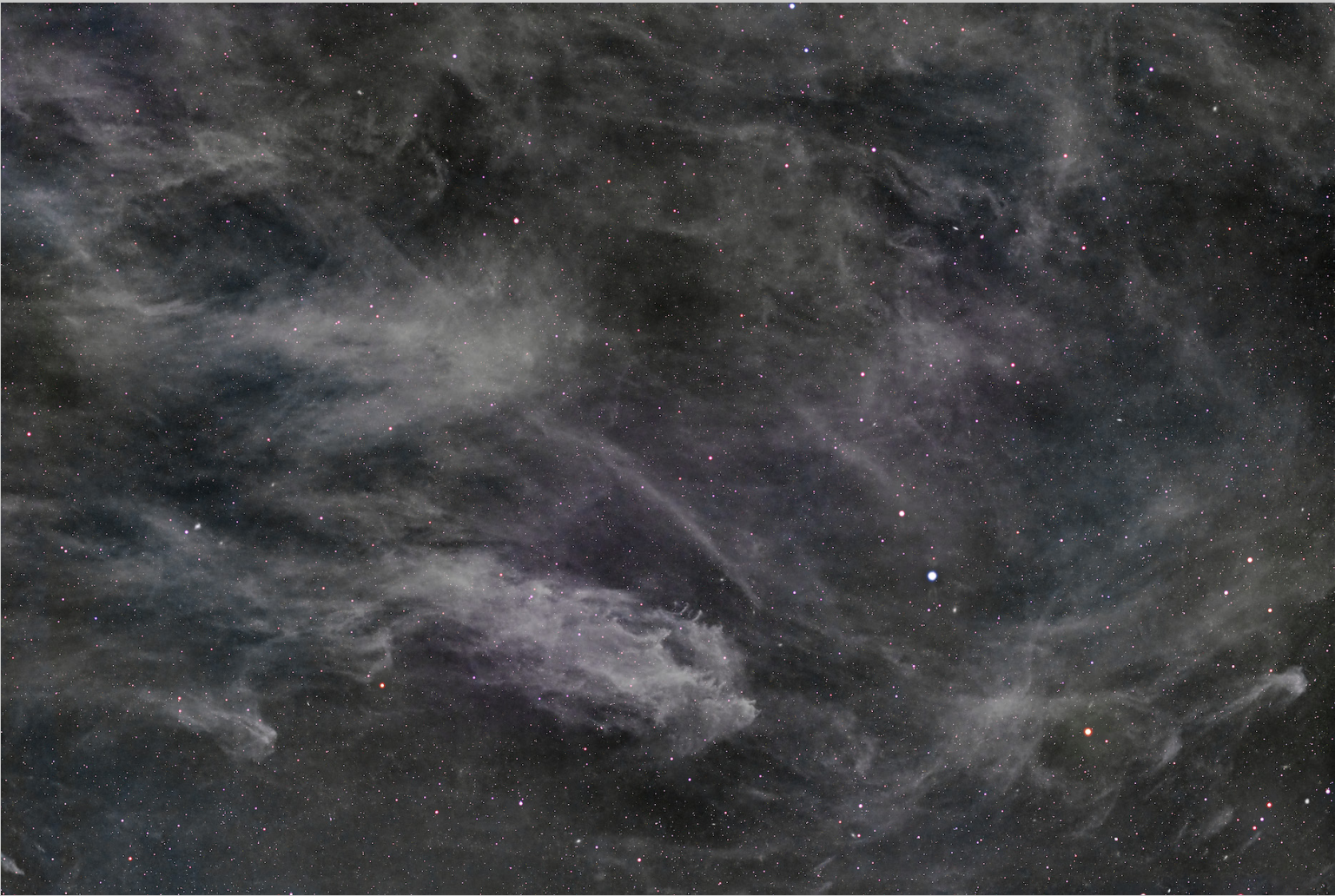
Join Chabot astronomers on the Observatory Deck for a free telescope viewing! Weather permitting, this is a chance to explore stars, planets and more through Chabot’s historic telescopes. Chabot’s three large historic telescopes offer a unique way to experience the awe and wonder of the Universe. Three observatory domes house the Center’s 8-inch (Leah, 1883) and 20-inch (Rachel, 1916) refracting telescopes, along with a 36-inch reflecting telescope (Nellie, 2003).

Are the skies clear for viewing tonight? Viewing can be impacted by rain, clouds, humidity and other weather conditions. Conditions can be unique to Chabot because of its unique location in Joaquin Miller Park. Before your visit, check out the Weather Station to see the current conditions at Chabot.

For more information, see <https://chabotspace.org/events/events-listing/> <https://eastbayastro.org/chabot-telescope-status/>



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WHATS UP

Adapted from Sky and Telescope

All times are Pacific Standard Time

July 2026

20	Mon	Facing southwest at dusk, see the Moon just $2\frac{3}{4}^{\circ}$ below Spica
21	Tue	Moon is at first quarter; Longest day of the year
24	Thu	At dusk see the moon just $2\frac{1}{2}^{\circ}$ lower lower left of Antares
29	Wed	Full Moon
30-31	Thu-Fri	All night see the Southern Delta Aquariid meteor shower

August 2026

6	Thu	Moon is at last quarter
7	Fri	In the morning, the crescent Moon climbs with the Pleiades
9	Sun	In the morning the Moon and Mars are side-by-side as they rise from the east-northeast
10	Mon	The moon forms a triangle with Caster and Pollux with the sides about being about 7°
11	Tue	At dawn look low on the east northeastern horizon to see the thin crescent Moon and Mercury less than 2° apart
12	Wed	New Moon
12-13	Wed-Thu	The Perseid meteor shower peaks
15	Sat	At dawn see the Moon leading Venus by about 4°
16	Sun	At dusk see the Moon about 6° apart from Spica

OFFICERS AND VOLUNTEER POSITIONS

Officers	Volunteer Positions	Newsletter	Potluck Coordinator
President Eric Dueltgen president@trivalleystargazers.org	Astronomical League Rep. Don Dossa alrep@trivalleystargazers.org	Scott Schneider (Editor) newsletter@trivalleystargazers.org	OPEN potluck@trivalleystargazers.org
Vice-President Aris Pope vice_president@trivalleystargazers.org	Club Star Party Coordinator Tom Dossa coordinator@trivalleystargazers.org	Webmaster Swaroop Shere webmaster@trivalleystargazers.org	Program Coordinator Ron Kane programs@trivalleystargazers.org
Treasurer Ernest Epley treasurer@trivalleystargazers.org	Del Valle Coordinator Dave Wiltzius delvalle@trivalleystargazers.org	Night Sky Network Rep. Ross Gaunt nnsn@trivalleystargazers.org	Publicity and Fundraising OPEN publicity@trivalleystargazers.org
Secretary Dave Lackey secretary@trivalleystargazers.org	Historian OPEN historian@trivalleystargazers.org	H2O Observatory Director / Rebuild Chairman Chuck Grant H2O@trivalleystargazers.org	Refreshment Coordinator OPEN
Past President Ron Kane past_president@trivalleystargazers.org	Librarian Ron Kane librarian@trivalleystargazers.org	Observing Program Coordinator Ron Kane awards@trivalleystargazers.org	Web and Email www.trivalleystargazers.org info@trivalleystargazers.org
	Loaner Scope Manager Ron Kane telescopes@trivalleystargazers.org	Outreach Coordinator Tom Dossa outreach@trivalleystargazers.org	TVS E-Group To Join the TVS E-Group just send an email to TVS at info@trivalleystargazers.org asking to join the group. Make sure you specify the email address you want to use to read and post to the group.

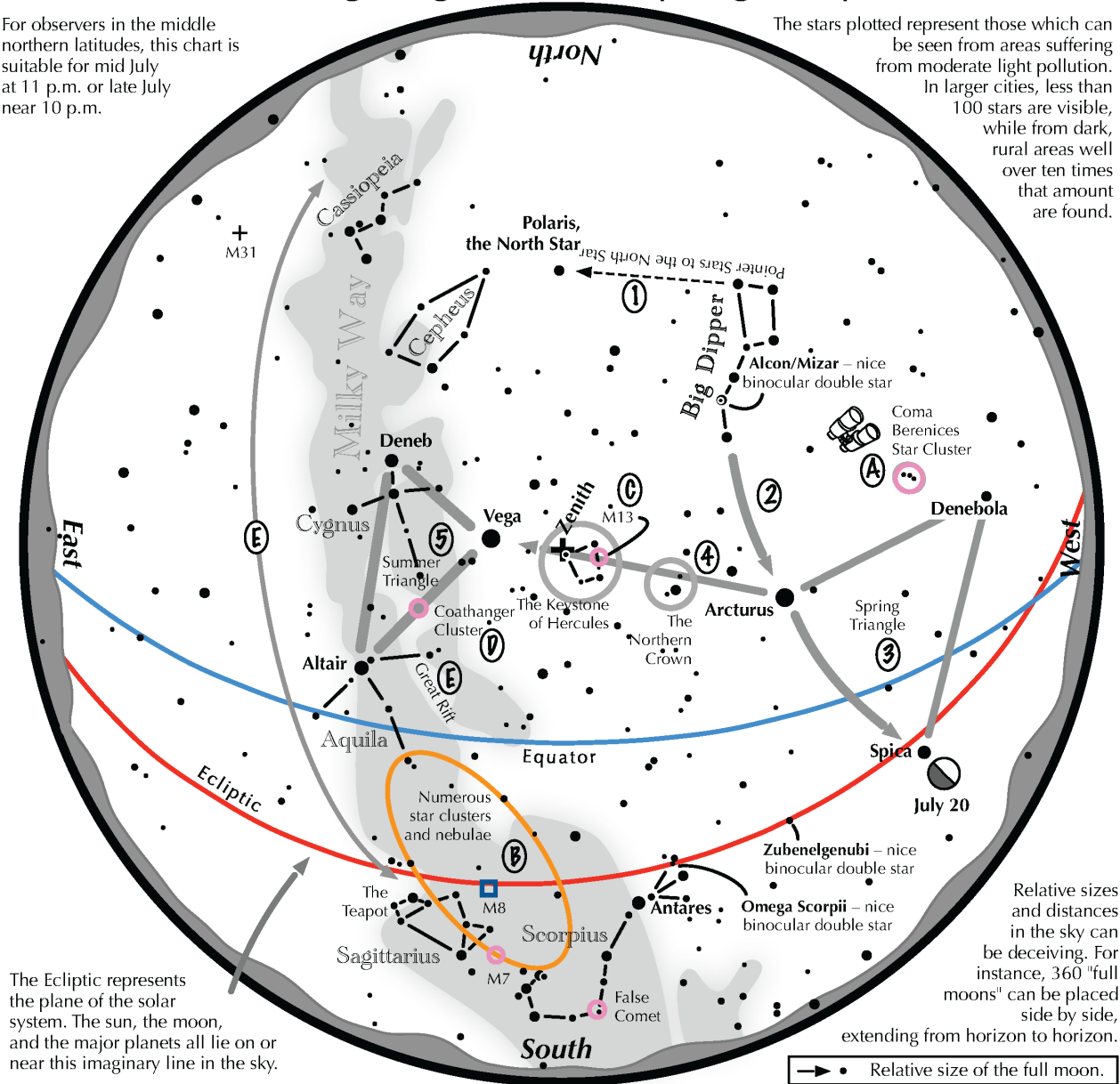
NAVIGATING THE NIGHT SKY FOR MONTH

Navigating the mid July Night Sky

2026

For observers in the middle northern latitudes, this chart is suitable for mid July at 11 p.m. or late July near 10 p.m.

The stars plotted represent those which can be seen from areas suffering from moderate light pollution. In larger cities, less than 100 stars are visible, while from dark, rural areas well over ten times that amount are found.



Navigating the mid July night sky: Simply start with what you know or with what you can easily find.

- 1 Extend a line north from the two stars at the tip of the Big Dipper's bowl. It passes by Polaris, the North Star.
- 2 Follow the arc of the Dipper's handle. It first intersects Arcturus, the brightest star in the July evening sky, then continues to Spica.
- 3 Arcturus, Spica, and Denebola form the Spring Triangle, a large equilateral triangle.
- 4 To the northeast of Arcturus shines another star of similar brightness, Vega. Draw a line from Arcturus to Vega. It first meets "The Northern Crown," then the "Keystone of Hercules." A dark sky is needed to see these two dim stellar configurations.
- 5 High in the East lies the Summer Triangle stars of Vega, Altair, and Deneb.

Binocular Highlights

- A: Between Denebola and the tip of the Big Dipper's handle, lie the stars of the Coma Berenices Star Cluster.
- B: Between the bright stars Antares and Altair, hides an area containing many star clusters and nebulae.
- C: On the western side of the Keystone glows the Great Hercules Cluster, containing nearly 1 million stars.
- D: 40% of the way between Altair and Vega, twinkles the "Coathanger," a group of stars outlining a coathanger.
- E: Sweep along the Milky Way for an astounding number of faint glows and dark bays, including the Great Rift.

Astronomical League www.astroleague.org; duplication is allowed and encouraged for all free distribution.



Tri-Valley Stargazers
P.O. Box 2476
Livermore, CA 94551
www.trivalleystargazers.org

Tri-Valley Stargazers Membership Application

Contact information:

Name: _____ Phone: _____

Street Address: _____

City, State, Zip: _____

Email Address: _____

Status (select one): _____ New member _____ Renewing or returning member

Membership category (select one): Membership term is for one calendar year, January through December.

_____ Student member (\$10). Must be a full-time high-school or college student.

_____ Regular member (\$30).

Hidden Hill Observatory Access (optional): Must be 18 or older.

_____ One-time key deposit (\$20). This is a refundable deposit for a key to H2O. New key holders must first hear an orientation lecture and sign a usage agreement form before using the observing site.

_____ Annual access fee (\$10). You must also be a key holder to access the site.

Donation (optional):

_____ Tax-deductible contribution to Tri-Valley Stargazers

Total enclosed: \$ _____

Member agrees to hold Tri-Valley Stargazers, and any cooperating organizations or landowners, harmless from all claims of liability for any injury or loss sustained at a TVS function. TVS will not share information with anyone except as detailed in our Privacy Policy (<http://www.trivalleystargazers.org/privacy.shtml>).

Mail this completed form along with a check to: Tri-Valley Stargazers, P.O. Box 2476, Livermore, CA 94551.